DEPARTMENT OF MATHEMATICS

MIDTERM TEST #2

MTH 240 – Calculus II

Last Name (Print): [redacted]. First Name: [redacted]. Student Number: [redacted]

Signature [redacted]

Date: March 15, 2013, 4:00 pm

Duration: 1.5 hours

Section (circle one)

C. Kim :	1 2 3 4
C. Grandison :	6 7 8 9
P. Lawrence :	11 12 13 14
A. Alvarez :	16 17 18 19 21 22 23 24

Instructions:

1. This is a closed-book test. **Notes, calculators and other aids are not permitted.**
2. Verify that your test has pages 1-7.
3. (a) Unless otherwise instructed, **make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.** Put a box around your final **answer.**

   (b) Write your solutions in the space provided. If you need more space, use the back of the page. Indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there. Marks (out of 50) are shown in brackets.
4. Do not separate the sheets.
5. Have your student card available on your desk.

For Instructor's use only.

Page	Mark
2	8 / 10
3	4 / 8
4	5 / 8
5	7 / 8
6	3 / 8
7	0 / 8
Total	27 /50

MECU

8/10

1. [10 marks] Determine whether each of the following series is convergent or divergent. If it is convergent find its value. All terms positive

Using Comparison Test.

$a_n = \frac{n^2}{\sqrt[3]{n^5+4n}}$

a) $\sum_{n=1}^{\infty} \frac{n^2}{\sqrt[3]{n^5+4n}}$

$b_n = \frac{n^2}{\sqrt[3]{n^5}}$

$b_n < a_n$

So if b_n diverges a_n also diverges.

Proving that b_n diverges

$\lim_{n\to\infty} \frac{n^2}{\sqrt[3]{n^5}}$

$= \lim_{n\to\infty} n^{2-5/3}$

$= \lim_{n\to\infty} n^{1/3} = \infty$

so as $n \to \infty$ $b_n \to \infty$ and as a consequence $a_n \to \infty$ and so given series is Divergent

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b) $\sum_{n=1}^{\infty} \frac{3}{2^{n+1}}$

$a_n = \frac{3}{2^{n+1}}$ its a p series of the form $\frac{1}{p^n}$

and in this case $n = n+1 > 1$ so CONVERGENT

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Find sum

$\sum_{n=1}^{\infty} \frac{3}{2^{n+1}} = 3 \sum_{n=1}^{\infty} \frac{1}{2^{n+1}}$

$= 3\left(\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} \cdots \frac{1}{2^{n+1}}\right)$

as $\lim_{n\to\infty} \left(\frac{1}{2^{n+1}}\right) \to 1$

$= 3(1) = 3$

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2. [8 marks] Determine whether the following series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$$

Checking if decreasing $f(n)$

$$y' = \frac{\frac{1}{n} n^2 - 2n \ln n}{n^4}$$

$$= \frac{n - 2n \ln n}{n^4}$$

$$= \frac{\cancel{n}(1 - 2\ln n)}{n^3}$$

will decrease when $2\ln n > 1$

$\ln n > \frac{1}{2}$

$n > e^{1/2}$

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Using Comparison Test.

$b_n = \frac{1}{n^2}$

$b_n < n$ when when $\ln n > 1$

so when $n > e$

$\lim_{n \to \infty} \frac{1}{n^2} = 0$ (Divergent test inconclusive as $=0$ could or could not be convergent.

Using limit comparison test.

$$\lim_{n \to \infty} \frac{\ln n}{n^2} \times \lim_{n \to \infty} \frac{n^2}{1} =$$

$$\lim_{n \to \infty} \left(\frac{\ln n}{n^2} \frac{n^2}{1} \right)$$ as $n \to \infty$ $\ln n \to \infty$ $\therefore \neq C$

$\therefore$ limit doesnt exist so divergent.

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3. [8 marks] Determine whether the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}(-1)^n\frac{(n!)^2}{(2n)!}$$

use ratio test.

$$a_{n+1} = (-1)^{n+1}\frac{((n+1)!)^2}{(2(n+1))!}$$

$$\frac{a_{n+1}}{a_n} = \left|\frac{((n+1)!)^{\cancel{2}}}{(\cancel{2}(n+1))!} \times \frac{(\cancel{2}n)!}{(n!)^2}\right|$$

(ignore -1^n ~~too~~ b/c its abs value)

$$\frac{a_{n+1}}{a_n} = \left|\frac{(n+1)!}{(n!)^2}\right| = \left|\frac{n!(n+1)}{(n!)^{\cancel{2}}}\right|$$

$$= \left|\frac{\cancel{(n+1)}}{(n-1)!\,\cancel{(n+1)}}\right|$$

$$= \lim_{n\to\infty}\left|\frac{1}{(n-1)!}\right| = 0$$

as $n \to \infty$ $(n-1)! \to \infty$ and $\frac{1}{n-1} \to 0$

$\therefore L = 0 < 1 \therefore$ convergent

4. [8 marks] Determine whether the series

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$$\sum_{n=1}^{\infty}(-1)^n \frac{1}{n+\sqrt{n}}$$

is absolutely convergent, conditionally convergent or divergent.

Using alternating series Test.

check i) $b_{n+1} \leq b_n$

ii) $\lim_{n\to\infty} b_n = 0$

$b_{n+1} \leq b_n$

$$\frac{1}{(n+1)+\sqrt{n+1}} \leq \frac{1}{n+\sqrt{n}}$$

This is true as "n+1" in denominator will always result in a smaller term overall as compared to having jus "n" in denominator.

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{\cancel{n}+\cancel{\sqrt{n}}} = 0$$

so by the alternating series test

the given series is convergent

checking with absolute comparison

$$\left|(-1)^n \frac{1}{n+\sqrt{n}}\right| = \frac{1}{n+\sqrt{n}}$$

$$b_n = \frac{1}{n}$$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty}\left(\frac{1}{n+\sqrt{n}} \times \frac{n}{1}\right)$$

$$= \lim_{n\to\infty}\left(\frac{n}{n+\sqrt{n}}\right)$$

$$= \lim_{n\to\infty}\left(\frac{\cancel{n}\ 1}{\cancel{n}(1+\sqrt{1/n})}\right)$$

$$= \lim_{n\to\infty} \frac{1}{1+\sqrt{1/n}}$$

$$= \frac{1}{1+0} = 1$$

QED

5. [8 marks] Find the orthogonal trajectories of the family of curves.

$$y = \frac{x}{1+kx}$$

$y = \frac{x}{1+kx}$

$1+kx = \frac{x}{y}$

$k = \left(\frac{x}{y} - 1\right)\frac{1}{x}$

$k = \left(\frac{1}{y} - \frac{1}{x}\right)$

$y = \frac{x}{1+kx}$

$y' = \frac{(1+kx) - (x(1+k))}{(1+kx)^2}$

$= \frac{1 + \cancel{kx} - x - \cancel{kx}}{(1+kx)^2}$

$m_1 = y' = \frac{1-x}{(1+kx)^2}$

$m_1 = \frac{1-x}{\left(1 + \left(\frac{1}{y} - \frac{1}{x}\right)x\right)^2}$

$= \frac{1-x}{\left(\cancel{1} + \frac{x}{y} - \cancel{1}\right)^2}$

$= \frac{(1-x)y^2}{x^2}$

$m_1 = \frac{y^2}{x^2} - \frac{y^2}{x}$

$\frac{dy}{dx} = y^2\left(\frac{1}{x^2} - \frac{1}{x}\right)$

$m_2 = \frac{x^2}{y^2} - \frac{x}{y^2}$

$m_2 = \frac{x^2 - x}{y^2} = \frac{dy}{dx}$

$dy\, y^2 = x^2 - x\, dx$

$\therefore y^2 = x(x-1)$

$m_1 \times m_2 = -1$

$\frac{y^2}{x^2} - \frac{y^2}{x} \times \frac{1}{\frac{y^2}{x^2} - \frac{y^2}{x}} = -1$

$= \frac{1}{\frac{y^2}{x^2} - \frac{y^2}{x}} = -1$

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$\sin x \to \cos x$

$\cos x \to -\sin x$

$\sin x = -\cos x$

6. [8 marks] Solve the initial-value problem.

isolating x terms from y terms.

$$xy' = y + x^2 \sin x, \qquad y(\pi) = 0$$

$xy' = y + x^2 \sin x$

$\cancel{x}y' = \cancel{x}\left(\frac{y}{x} + x^{\cancel{2}} \sin x\right)$

$\frac{dy}{dx} \cancel{\frac{x}{y}} = \cancel{x} \sin x$

$\frac{dy}{dx} \frac{1}{y} = \sin x$

$\frac{1}{y} dy = \sin x \, dx$

$\int \frac{1}{y} dy = \int \sin x \, dx$

$\ln|y| = -\cos x + C_1$

$e^{\ln|y|} = e^{-\cos x + C_1}$

$|y| = e^{-\cos x + C_1}$

$y = \pm e^{-\cos x}(e^{C_1})$

$e^{C_1} = K$

$y = K e^{-\cos x}$

$y(\pi) = 0$

$y(\pi) = Ke^{-\cos \pi} = 0$

$y(\pi) = Ke^{-0} = 0$

$K \cdot 1 = 0$

$K = 0$

$\cos \pi = 0$

so $y = 0 \cdot e^{-\cos x}$